Introduction to Robotics

1. What is a Robot?

A robot is a programmable machine capable of carrying out a series of actions autonomously or semi-autonomously. It typically consists of a mechanical structure, sensors, actuators, and a control system.

2. Key Components

Sensors: gather information about the environment (e.g. ultrasonic, infrared, cameras). Actuators: convert control signals into physical motion (e.g. servo motors, stepper motors). Controller: the processing unit (microcontroller or onboard computer) that runs the control logic. End effector: the tool at the end of a robotic arm, such as a gripper.

3. Degrees of Freedom (DOF)

Degrees of freedom refers to the number of independent movements a robot can perform. A robotic arm with 6 DOF can position its end effector at any point in 3D space with any orientation, similar to a human arm.

4. Types of Robots

Industrial robots: used in manufacturing for tasks like welding and assembly. Mobile robots: move through their environment, e.g. warehouse robots or drones. Humanoid robots: designed to resemble and mimic human movement. Autonomous vehicles: robots that navigate without human control, used in cars and delivery bots.

5. Control Systems

Open-loop control executes a predefined sequence without feedback. Closed-loop control uses sensor feedback to continuously adjust actions, allowing the robot to correct errors in real time. Most modern robots use closed-loop control for precision tasks.